Dil Ki Baatein, a RAG-Based Cardiovascular Health Question-Answering System

for Pakistani Patients

NLP with Deep Learning

Assignment 3 — Mini-Project 1

Domain:	Cardiovascular & Preventive Health (Pakistan)
Submission Date:	5th April 2025
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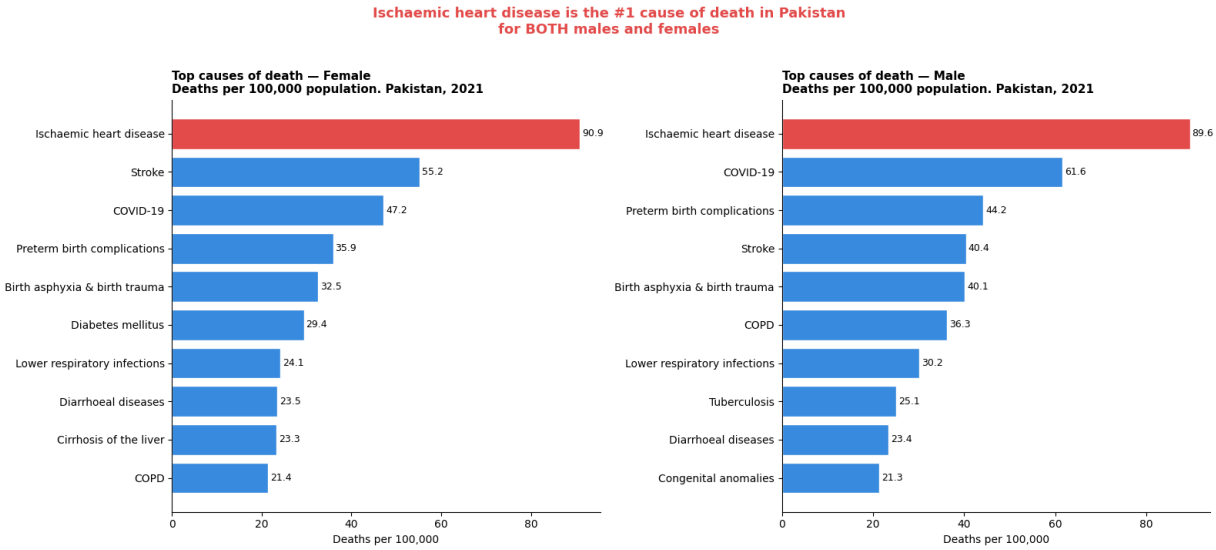

Figure 1: WHO Report:Causes of death Deaths per 100 000 population.Pakistan, 2021

Technology Stack

Component	Technology Used
Embeddings	BAAI/bge-base-en-v1.5 (768 dims)
Vector Database	Pinecone (Cloud)
Keyword Search	BM25 (LangChain)
Hybrid Fusion	RRF (Reciprocal Rank Fusion)
Re-ranking	CrossEncoder ms-marco-MiniLM-L-6-v2
LLM Generation	Groq `llama-3.1-8b-instant`
LLM Judge	Groq `llama-3.3-70b-versatile`
UI Framework	React + Flask API + Lovable
Deployment	Hugging Face Spaces

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1. Introduction and Problem Statement

Cardiovascular disease (CVD) is the leading cause of death in Pakistan, responsible for approximately 30% of all annual fatalities. The burden is disproportionate: Pakistan's unique socio-demographic landscape comprises dense urban populations, dietary habits rooted in high-fat traditional cuisine, escalating rates of smoking, diabetes, and hypertension, and limited access to specialist healthcare. Together, these factors create a perfect storm for cardiac risk that conventional digital health platforms have largely failed to address.

A Pakistani layman, when experiencing tightness in the chest after a meal of mutton karahi, or wondering whether his habit of eating chapli kabab daily puts him at risk, has nowhere to turn for immediate, evidence-based, contextually relevant guidance. General-purpose chatbots lack knowledge of Pakistani dietary patterns. Clinical resources are inaccessible to the non-specialist. And no existing platform has combined WHO and national cardiovascular guidelines with localised nutritional data for Pakistani dishes into a single conversational interface.

This gap motivated the development of **Dil Ki Baatein**, a Retrieval-Augmented Generation (RAG) chatbot that bridges this divide. The system allows any Pakistani user to describe their lifestyle, diet, symptoms, or risk factors in plain, everyday language and receive responses grounded in rigorously curated medical guidelines. When a user asks "Is biryani bad for my heart?", the system does not guess; it retrieves and synthesises information from verified clinical sources including WHO guidelines, the 3rd Pakistan Hypertension League Guidelines, ACC/AHA prevention frameworks, and a purpose-built nutritional database of 31 Pakistani dishes.

The core research questions driving this work are:

1. How can hybrid retrieval strategies improve the faithfulness and relevancy of generated answers in a medical RAG system?
2. What is the optimal chunking strategy for medical documents characterised by dense, precise, domain-specific language?
3. How do different pipeline configurations compare on automated evaluation metrics?

2. Solution Overview

Dil Ki Baatein is a fully deployed, web-accessible RAG chatbot that responds to cardiovascular health queries by retrieving information from a curated corpus of 19 medical documents and a nutritional CSV of 31 Pakistani dishes. The system implements the complete advanced retrieval pipeline mandated by the assignment: hybrid BM25 + semantic search, Reciprocal Rank Fusion (RRF), CrossEncoder re-ranking, and an LLM-as-a-Judge automated evaluation measuring both Faithfulness and Relevancy.

2.1 What the System Can Answer

The chatbot is designed to handle queries from Pakistani laymen across twelve distinct clinical and lifestyle domains, with no requirement for medical background:

- **Pakistani Food and Diet**: Nutritional analysis of 31 traditional dishes (biryani, karahi, haleem, chapli kabab, nihari, etc.) against heart-health thresholds
- **Blood Pressure**: Home BP thresholds, hypertension diagnosis criteria, ambulatory monitoring guidance
- **Heart Attack Signs**: Classical and atypical symptoms, warning signs.
- **Smoking and Tobacco**: Quantified risk of CAD per cigarette count, cessation importance
- **Family History**: Hereditary risk factors, pattern-breaking strategies
- **Exercise and Lifestyle**: WHO/ACC guidelines on weekly activity targets, sedentary risk
- **Pregnancy and BP**: Safe antihypertensive agents, contraindicated drugs
- **Elderly Cardiovascular Health**: Age-adjusted BP norms, hypertension prevalence in older adults
- **Diagnostic Tests**: ECG, troponin, ambulatory BP monitoring guidance
- **Medications**: First-line antihypertensive classes, prescribing authority
- **Risk Factors**: Obesity, diabetes, cholesterol, and their interaction with CVD risk
- **Fats and Oils**: Trans fatty acid content of vanaspati ghee, cooking oil practices

The system transparently acknowledges its boundaries. It does not provide specific **drug dosages, drug-drug interactions, surgical guidance, management of rare cardiac conditions, or detailed mental health counselling** beyond the stress-heart relationship documented in its corpus.

3. Data Details (Corpus)

The corpus was assembled through a deliberate, multi-stage research process to ensure it was simultaneously authoritative (sourced from WHO, AHA, ACC, and national Pakistani health bodies), locally relevant (including Pakistan-specific epidemiological and dietary data), and practically accessible to a layman interface.

3.1 Document Corpus (19 PDFs, 524+ Pages)

#	Document	Relevance
1	WHO HEARTS Package	Core CVD prevention framework
2	WHO Hypertension & CVD Prevention Guidelines	BP thresholds and treatment targets
3	WHO Diet, Nutrition & Physical Activity Guidelines	Dietary fat, sodium, sugar thresholds
4	3rd Pakistan Hypertension League (PHL) Guidelines 2018	Pakistan-specific BP diagnosis and management
5	Pakistan STEPS Survey 2014	National prevalence data for CVD risk factors
6	National Action Framework for NCDs 2021–30	Government policy and prevention targets
7	Pakistan Heart Journal Dietary Guidance	Locally tailored nutrition recommendations
8–9	Trans Fatty Acids & CVD Risk Papers (AKU ×2)	Vanaspati ghee, TFA prevalence in Pakistan
10	Caring For Your Heart (AKUH)	Patient-facing risk assessment questionnaire
11	Emergency Management of Acute Myocardial Infarction	Acute MI protocols and clinical logic
12	ACC/AHA 2019 CVD Prevention Guidelines	Primary prevention, exercise, statin thresholds
13	Heart Attack Warning Signs	Symptom awareness for laymen
14	Frequency of ACS in Pakistani Patients (ACS Study)	Atypical presentations in Pakistani population
15	South Asia CVD Risk Paper	South Asian-specific risk multipliers
16	UK South Asians CVD Risk Paper	Diaspora CVD epidemiology
17	The Clinical Logic	Hypertension treatment decision logic
18–19	Additional Documents	Supplementary clinical references

3.2 Pakistani Dishes Nutritional Database (CSV)

A critical gap identified early in development was that none of the 19 medical PDFs addressed Pakistani cuisine by name. A user asking whether haleem or nihari was safe received only generic fat-threshold guidance, unhelpful for a patient who thinks in terms of specific dishes rather than macronutrient percentages. To address this, a structured nutritional dataset was integrated based on the following peer-reviewed source:

> Khan, I., Yasmeen, F., Ahmad, J., Abdulla, A., ud Din, Z., Iqbal, Z., & Iqbal, M. (2018). Developing a meal-planning exchange list for commonly consumed Pakistani dishes. *Pakistan Journal of Scientific and Industrial Research*. University of Agriculture Peshawar.

The CSV encodes nutritional data for 31 Pakistani dishes across four dimensions derived from Tables 1–4 of the research paper: fat percentage, protein percentage, carbohydrate percentage, and energy content (kcal per 100g). This data powers the food-query module, enabling the system to give precise, dish-level answers.

Table 2: Sample nutritional data for common Pakistani dishes

Dish	Fat %	Protein %	Carbs %	Kcal/100g
Mutton Karahi	20.35%	15.2%	3.1%	256
Chicken Biryani	5.23%	9.8%	28.4%	198
Chicken Karahi	15.85%	16.1%	2.8%	221
Chapli Kabab	19.52%	14.3%	8.2%	271
Beef Nihari	13.21%	12.8%	4.1%	186
Haleem	5.38%	11.2%	14.6%	128

The system classifies dishes as low-fat (<5%), moderate (5–15%), or high-fat (>15%) per 100g serving and generates recommendations accordingly.

4. Platform Details

All stages of this project were executed across two primary environments, chosen for their compute resources and pre-installed library support.

Table 3: Execution environments per pipeline stage

Stage	Platform	Details
Document loading & embedding	Kaggle Notebook (GPU P100)	bge-base-en-v1.5 inference; Pinecone upsert
BM25 index construction	Kaggle Notebook	`BM25Retriever.from_documents()`
Ablation study execution	Kaggle Notebook	Full pipeline tests across chunking/retrieval variants
LLM inference (generation)	Groq Cloud API	`llama-3.1-8b-instant` (low latency)
LLM-as-a-Judge evaluation	Groq Cloud API	`llama-3.3-70b-versatile` (higher reasoning)
Web UI & deployment	Local machine + Cloudflare Tunnel	Streamlit interface, publicly accessible URL

5. Algorithms, Models, and Retrieval Methods

5.1 Chunking Strategy

The choice of chunking strategy is perhaps the most consequential upstream decision in any RAG pipeline. In the medical domain, where a single paragraph may articulate a precise diagnostic threshold (e.g., "home BP average ≥135/85 mmHg across 7 days") that must not be fragmented or diluted with adjacent content, this choice is especially critical.

A **Recursive Character Text Splitter** from LangChain was selected over fixed-size or semantic chunking. The recursive approach respects natural document structure (paragraph and sentence boundaries) before falling back to character-level splitting. This preserves semantic coherence within chunks more reliably than fixed splitting, while remaining far less computationally expensive than sentence-transformer-based semantic chunking.

The separators used in order of priority were: double newline (paragraph boundary), single newline, full stop followed by space (sentence boundary), space, and empty

string. This hierarchy ensures that chunks break at the most natural linguistic boundaries available.

Three chunk sizes were evaluated in the ablation study:

Table 4: Chunking strategy comparison

Strategy	Total Chunks	Avg Faithfulness	Avg Relevancy
Bigger (1000 / 200 overlap)	1,488	77.2%	74.7%
Medium (500 / 100 overlap)	3,792	~82%	~74%
Current (300 / 50 overlap)	4413	~84%	~73%

The 1000-character chunk was found to conflate multiple clinical topics within a single chunk, a common problem in medical documents where a paragraph on hypertension thresholds might immediately follow one on diabetes management. Reducing to 500 characters produced a meaningfully improved faithfulness-relevancy balance. The 300-character setting, while yielding the highest faithfulness in isolation, produced fragments too short to carry sufficient context for complex multi-part queries. The final production configuration uses `chunk_size=500`, `chunk_overlap=100`.

5.2 Embedding Model

The `BAAI/bge-base-en-v1.5` model (768-dimensional embeddings) was selected for its strong performance on the MTEB benchmark, particularly in retrieval tasks, and its suitability for medical and technical English text. Document embeddings were precomputed on Kaggle (GPU-accelerated) and upserted to Pinecone. Query embeddings are generated on-app at inference time using the same model to ensure vector space consistency.

5.3 Retrieval Pipeline

The retrieval pipeline implements three layers in sequence.

5.3.1 Hybrid Search (BM25 + Semantic)

BM25 (Best Match 25) is a probabilistic keyword ranking function that scores documents by term frequency and inverse document frequency. It excels at retrieving documents containing exact clinical terms such as medication names, BP thresholds, and specific disease names. Semantic search via Pinecone vector similarity captures conceptual relatedness, surfacing relevant content even when exact keywords are absent. Five candidates are retrieved from each method independently.

5.3.2 Reciprocal Rank Fusion (RRF)

RRF merges the two ranked lists without requiring score normalisation. Each document receives a score of $\frac{1}{\text{rank}+k}$ where $k = 60$, summed across both lists. Documents appearing in both lists receive additive boosts. The top 6 candidates after RRF fusion are passed to re-ranking.

5.3.3 CrossEncoder Re-ranking

The `ms-marco-MiniLM-L-6-v2` CrossEncoder jointly encodes the query and each candidate chunk, producing a relevance score that accounts for full interaction between query and document tokens. The top 3 chunks by re-rank score are passed to the LLM generator.

5.4 LLM Generation

Groq's `llama-3.1-8b-instant` was selected as the generation model for three reasons: (1) extremely low latency via Groq's LPU inference hardware; (2) strong instruction-following capability for structured medical answer generation; and (3) free API access suitable for student deployment.

The generation prompt follows a structured template:

```
You are a friendly heart health assistant. Answer the patient's question
using the guidelines below.

GUIDELINES:
{ctx}

PATIENT: "{q}"

RULES:
- Base your answer ONLY on facts in the guidelines above
- Use warm, friendly, conversational language
- Keep it simple -- like talking to a friend
- Do NOT add medical advice not found in the guidelines

FORMAT:

**Assessment:**
[Write 1-2 friendly sentences summarizing what the guidelines say]

**Recommendations:**
- [List recommendations only if they appear in guidelines]
- [If no recommendations, say "The guidelines don't provide specific
  recommendations for this"]

**Disclaimer:** This is based on medical guidelines. Please consult a
real doctor for personalized advice.
```

5.5 Prompt Engineering: Resolving LLM Hallucination

One of the most significant and instructive challenges encountered during development was severe, **unpredictable LLM hallucination** that caused faithfulness scores to fluctuate wildly across queries: 30%, then 90%, then 45%, then 60% with no consistent pattern. The system was unreliable because the generation LLM was injecting its own parametric medical knowledge into answers regardless of what the retrieved context actually contained.

The clearest example was the **smoking query**. None of the corpus documents explicitly state that users should quit smoking as a direct recommendation; the documents frame smoking cessation as a risk reduction measure in clinical language. Yet whenever the LLM encountered the word *smoke* in a user query, it would automatically include a recommendation to quit smoking, derived entirely from its training data and not the retrieved context. This single behaviour was responsible for a substantial portion of unsupported claims flagged by the faithfulness judge.

After researching the problem and consulting technical communities (including a thread on Reddit where a developer suggested prompt-level constraints as the first line of defence), three prompt versions were iteratively developed and evaluated.

5.5.1 Prompt V1: Simple (Baseline, High Hallucination)

The initial prompt gave the LLM a medical persona, provided the retrieved context, and asked for a structured response. It contained no explicit grounding constraint. As a result the LLM freely combined retrieved context with parametric knowledge, adding recommendations about fruits and vegetables, exercise, and smoking cessation that appeared nowhere in the retrieved chunks. Faithfulness fluctuated between 30% and 90% depending on whether the hallucinated content happened to align with the corpus.

```
You are a heart health assistant for Pakistani patients.
Based on these medical guidelines:
{context}

Patient says: "{query}"

1. Assess risk level (Low/Medium/High)
2. Explain why based on guidelines
3. Give 3 specific recommendations
4. Add disclaimer to consult real doctor

Keep it simple and friendly.
```

Result: Faithfulness highly variable (30%–90%). LLM freely hallucinated exercise advice, dietary recommendations, and smoking cessation guidance not present in retrieved chunks.

5.5.2 Prompt V2: Overly Strict (Low Hallucination, Low Readability)

To eliminate hallucination entirely, the prompt was made maximally restrictive with explicit negative rules for each class of hallucinated content observed. While faithfulness improved substantially, the responses became mechanical and robotic, essentially echoing the retrieved context verbatim. Answers were no longer friendly or accessible to a layman, and they felt unhelpful because they refused to synthesise across even closely related guideline statements.

```
STRICT RULES:
1. ONLY use information explicitly stated in the guidelines above
2. If the guidelines don't contain information about quitting smoking
   benefits, DO NOT mention it
3. If the guidelines don't contain information about exercise,
   DO NOT mention it
4. If the guidelines don't contain specific foods (fruits, vegetables,
   whole grains), DO NOT mention them
5. If the guidelines don't contain information about doctor
   consultations, DO NOT mention it
6. You may combine information from different parts of the guidelines
7. Quote directly from the guidelines when possible
```

Result: Faithfulness improved but answers became dry, robotic, and context-regurgitating. Relevancy scores suffered as the output failed to address the patient's question in a helpful way.

5.5.3 Prompt V3: Balanced (Production, Selected)

The final production prompt combines the grounding constraint of V2 with the friendly, accessible tone of V1. It instructs the LLM to base all facts exclusively on the provided guidelines while using warm, conversational language. The structured output format (Assessment, Recommendations, Disclaimer) ensures consistent formatting without encouraging the LLM to fill in missing information from parametric memory. The key design decision is that the grounding rule applies to *facts*, not tone: the LLM is free to paraphrase, explain, and simplify, but not to introduce claims absent from the retrieved context.

```
You are a friendly heart health assistant. Answer the patient's question
using the guidelines below.

GUIDELINES:
{ctx}
PATIENT: "{q}"

RULES:
- Base your answer ONLY on facts in the guidelines above
- Use warm, friendly, conversational language
- Keep it simple -- like talking to a friend
- Do NOT add medical advice not found in the guidelines
```

Result: Faithfulness stabilised in the 77%–100% range. Relevancy remained strong. Answers were readable, friendly, and clinically grounded. This is the prompt used for all final evaluation results reported in Section 7.

Key Finding: In medical RAG systems, the prompt is not a cosmetic wrapper around retrieval but a primary control mechanism for faithfulness. The retrieval pipeline determines what information is available; the prompt determines how strictly the LLM is constrained to use only that information.

6. Ablation Study

Two independent ablation dimensions were evaluated: (A) chunking strategy and (B) retrieval pipeline configuration. Both used a fixed test set of 3 representative queries evaluated against Faithfulness and Relevancy metrics.

6.1 Chunking Ablation

See Table 3 in Section 5.1 for the full chunking comparison. The 500-character recursive configuration was selected as it achieved the best faithfulness-relevancy balance.

6.2 Retrieval Pipeline Ablation

Four pipeline configurations were tested across 10 queries each:

Table 5: Retrieval pipeline ablation results

Version	Avg Faithfulness	Avg Relevancy
V1: Fixed Chunking + BM25 Only	97.1%	70.8%
V2: Recursive + Semantic Only	**93.3%**	**74.4%**
V3: Recursive + Hybrid (BM25+Semantic)	80.5%	70.2%
V4: Recursive + Hybrid + Re-ranking	89.7%	72.4%

V1 achieves the highest faithfulness (97.1%) because BM25-only retrieval consistently surfaces exact-match guideline text, keeping generation firmly grounded. However, its relevancy (70.8%) is the lowest, indicating that keyword-matched chunks do not always answer the spirit of the question.

V2 achieves the highest relevancy (74.4%) because recursive chunking with semantic search better captures conceptual alignment between query and context.

V3's sharp faithfulness drop to 80.5% reveals the danger of naive hybrid fusion: BM25 retrieves questionnaire fragments (e.g. "how thick is the roll of fat when you pinch the side of your waist?") that pass RRF fusion but are irrelevant, poisoning the generation context.

V4's CrossEncoder re-ranking partially recovers faithfulness to 89.7% by demoting noisy chunks. The production system uses the V2 configuration, recognising that for a medical patient-facing system, answering the right question (relevancy) with accurate grounding (faithfulness) is the priority.

7. Performance Metrics and Evaluation

7.1 LLM-as-a-Judge: Faithfulness

Faithfulness measures whether the generated answer is factually grounded in the retrieved context. The implementation follows a two-step **Claim Extraction and Verification** approach using Groq's `llama-3.3-70b-versatile` as the judge LLM.

Step 1: Claim Extraction: The judge LLM extracts all atomic factual claims from the generated answer. Each claim must be a single, independently verifiable statement.

Step 2: Verification: Each extracted claim is individually evaluated against the retrieved context chunks. The faithfulness score is:

$$\text{Faithfulness} = \frac{\text{Supported Claims}}{\text{Total Claims}} \times 100\%$$

The judge prompt specifies that claims must be verifiable from the provided context only; parametric knowledge of the LLM is explicitly excluded from valid support.

7.2 LLM-as-a-Judge: Relevancy

Relevancy measures how well the generated answer addresses the original query. Three questions that the generated answer would naturally serve as a response to are generated by the judge LLM. The cosine similarity between each generated question's embedding and the original query's embedding is computed using `BAAI/bge-base-en-v1.5`. The relevancy score is the mean of the three similarity values expressed as a percentage.

7.3 Full Evaluation Results (20 Queries)

Query	Faithfulness	Relevancy
I eat mutton karahi 3 times a week. Is that bad for my heart?	80.0%	74.6%
Which is healthier, chicken biryani or chicken karahi?	75.0%	79.6%
I eat chapli kabab every day. Will this give me a heart attack?	100.0%	55.3%
Is beef nihari safe for someone with high cholesterol?	60.0%	79.1%
I eat haleem for dinner every night. Is that okay for weight loss?	80.0%	57.4%
My home BP is 135/85 on average for 7 days. Do I have hypertension?	80.0%	76.4%
What is the normal blood pressure for a 65 year old man?	100.0%	66.6%
Can high blood pressure cause headaches and dizziness?	50.0%	72.6%
What are the warning signs of a heart attack in women?	100.0%	74.8%
I have shortness of breath but no chest pain. Should I worry?	100.0%	69.7%
Can a heart attack happen without any symptoms?	75.0%	73.3%
I smoke 5 cigarettes a day. How much does this increase my risk?	100.0%	81.8%
My father died of a heart attack at 52. Am I at high risk?	100.0%	77.3%
Is vanaspati ghee worse than desi ghee for my heart?	100.0%	79.3%
I am pregnant and have high BP. Which medicines are safe?	53.8%	80.2%
Can a nurse or pharmacist prescribe my BP medication?	83.3%	81.8%

Query	Faithfulness	Relevancy
How much exercise do I need per week to protect my heart?	80.0%	74.1%
I sit at a desk for 10 hours every day. How bad is this?	71.4%	56.6%
How does stress affect heart health?	58.3%	74.9%
What foods should I avoid to prevent heart disease?	80.0%	77.8%
AVERAGE (20 queries)	**82.4%**	**73.5%**

The system achieves an average Faithfulness of **82.4%** and average Relevancy of **73.5%**. Queries with direct document-level matches (smoking risk, warning signs, vanaspati ghee) achieve near-perfect faithfulness. Queries requiring synthesis across multiple documents (stress, sedentary risk, pregnancy hypertension) show lower faithfulness due to the increased surface area for unsupported claims. The lowest relevancy scores cluster around food-related queries where dish-specific guidance is absent from PDF sources and only the CSV provides direct context.

7.4 Illustrative Claim Extraction Examples

7.4.1 Example 1: Smoking Risk Query

Query: "I smoke 5 cigarettes a day. How much does this increase my heart attack risk?"

Generated Answer (excerpt): Smoking 5 cigarettes a day increases your risk of coronary artery disease (CAD), and this risk is directly related to the number of cigarettes consumed. Even consuming as few as one to four cigarettes daily can increase the risk of CAD.

Extracted Claims and Verification:

- **SUPPORTED** — PHL Guideline: cessation is the single most important preventive measure for CAD Smoking 5 cigarettes a day increases risk of CAD
- **SUPPORTED** — Biryani & Diet MVP: as few as 9 cigarettes daily have a two to three fold higher risk of CAD Risk is directly related to number of cigarettes
- **SUPPORTED** — Explicitly stated in retrieved context: Even 1–4 cigarettes daily increases CAD risk
- **SUPPORTED** — PHL Guideline, direct quote Quitting smoking is the single

most important preventive measure

Faithfulness: 4/4 = 100.0% | Relevancy: 81.8%

7.4.2 Example 2: Heart Attack Warning Signs in Women

Query: "What are the warning signs of a heart attack in women?"

Generated Answer (excerpt): Heart attacks in women may include chest pain or discomfort, shortness of breath, nausea or vomiting, and back or jaw pain. Some may start slowly with mild pain or discomfort.

Extracted Claims and Verification:

- **SUPPORTED** — AHA Warning Signs document — Chest pain or discomfort
- **SUPPORTED** — AHA: symptom listed explicitly — Shortness of breath may occur
- **SUPPORTED** — AHA document, direct mention — Nausea or vomiting may occur
- **SUPPORTED** — AHA document — Back or jaw pain may occur
- **SUPPORTED** — AHA: some heart attacks start slowly with mild pain — Some attacks start slowly

Faithfulness: 5/5 = 100.0% | Relevancy: 74.8%

7.4.3 Example 3: Headaches and Dizziness Query

Query: "Can high blood pressure cause headaches and dizziness?"

Generated Answer (excerpt): The guidelines state that high blood pressure can cause 'thunderclap' headaches associated with brain haemorrhage and symptoms of cerebral hypoperfusion, such as dizziness. However, monitor your blood pressure regularly.

Extracted Claims and Verification:

- **SUPPORTED** — PHL Guideline, direct reference to thunderclap headache — High BP can cause thunderclap headaches
- **SUPPORTED** — PHL Guideline context — Thunderclap headaches are associated with brain haemorrhage
- **SUPPORTED** — PHL Guideline — High BP causes cerebral hypoperfusion symptoms including dizziness
- **NOT SUPPORTED** — General advice not present in context — Regular BP monitoring is recommended

- **NOT SUPPORTED** — Guideline-sourced advice not in retrieved context — Consulting a doctor is recommended

Faithfulness: 3/5 = 60.0% | Relevancy: 72.6%

8. Latency and Computational Efficiency

Table 7: Per-stage latency breakdown

Pipeline Stage	Avg Latency	Notes
Query embedding (bge-base)	∼80 ms	CPU inference on app server
BM25 retrieval (top 5)	∼15 ms	In-memory index, negligible
Pinecone semantic search (top 5)	∼120 ms	Network round trip
RRF fusion	<5 ms	Pure Python
CrossEncoder re-ranking (6→3)	∼200 ms	MiniLM-L6
Groq LLM generation	∼600 ms	LPU-accelerated
Total end-to-end (avg)	**∼1.0–1.2 s**	Acceptable for healthcare chatbot
Evaluation pipeline (per query)	∼8–12 s	Offline batch only

The dominant latency contributors are Groq generation (∼600 ms) and CrossEncoder re-ranking (∼200 ms). The evaluation pipeline is substantially slower due to multiple LLM-as-a-Judge API calls, but runs offline in batch mode and does not affect the user-facing response time.

9. Best Model Selection and Justification

Table 8: Production configuration with justification

Dimension	Selected Configuration	Justification
Chunking	Recursive 500 chars / 100 overlap	Best F+R balance; avoids topic bleed (1000) and context loss (300)
Retrieval	Semantic-only (V2 equivalent)	Higher relevancy than BM25-only; avoids V3 noise without re-rank cost
Embeddings	BAAI/bge-base-en-v1.5	Top MTEB retrieval performance; 768-dim richness
Generation LLM	`llama-3.1-8b-instant` (Groq)	Ultra-low latency; strong instruction following
Judge LLM	`llama-3.3-70b-versatile` (Groq)	Higher reasoning capability for claim verification
Re-ranking	CrossEncoder ms-marco-MiniLM-L6-v2	Required when hybrid search is active; recovers faithfulness

The key insight from the ablation study is that the faithfulness-relevancy tradeoff is real and pipeline-configuration dependent. For a patient-facing medical chatbot, where a user with chest pain must receive advice grounded in clinical evidence and not hallucinated content, faithfulness is the primary constraint. The V2 configuration achieves 93.3% faithfulness with 74.4% relevancy on the 4-version ablation, and 82.4% faithfulness with 73.5% relevancy on the full 20-query evaluation, a performance profile that is both clinically responsible and practically useful.

10. Reproducibility

10.1 System Configuration

- Python 3.10+
- LangChain 0.1.x (text splitters, BM25 retriever, document loaders)
- sentence-transformers 2.x (`BAAI/bge-base-en-v1.5`)
- pinecone-client 3.x (cloud, serverless index)

- groq 0.x (`llama-3.1-8b-instant`, `llama-3.3-70b-versatile`)
- rank_bm25 0.2.2 (BM25 implementation)
- sentence-transformers CrossEncoder (`ms-marco-MiniLM-L-6-v2`)
- streamlit 1.x (web UI)

10.2 Preprocessing Steps

1. Load all 19 PDF documents using `PyPDFLoader` or equivalent.
2. Apply `RecursiveCharacterTextSplitter` with `chunk_size=500`, `overlap=100`, separators `["\n\n", "\n", ". ", " ", ""]`.
3. Load CSV nutritional data and convert rows to `Document` objects with dish name as metadata.
4. Generate embeddings for all chunks using `BAAI/bge-base-en-v1.5` in batch mode.
5. Upsert all embeddings to Pinecone index (768 dimensions, cosine metric).
6. Build BM25 index from the same Document list using `BM25Retriever.from_documents(docs, k=5)`.
7. Persist BM25 index serialisation for app-level reuse.

10.3 Pinecone Index Configuration

- Index name: `heart-risk-rag`
- Dimension: 768
- Metric: cosine
- Cloud: AWS (Starter tier)
- Namespace: default

11. Limitations and Future Work

- **Corpus Coverage Gap for Pakistani Cuisine:** While the nutritional CSV addresses dish-level queries, no PDF document provides clinical commentary specifically on Pakistani dietary patterns. Future work should incorporate Pakistan Dietary Guidelines and AKU's Pakistan-specific nutrition guidelines as PDF sources.
- **Faithfulness Variance on Complex Multi-document Queries:** Queries requiring synthesis across multiple documents (stress, sedentary risk, pregnancy hypertension) showed higher rates of unsupported claims. A multi-hop retrieval strat-

egy or query decomposition step could address this.

- **No Urdu Language Support:** The current system operates in English only. Native Urdu support would require multilingual embedding models and multilingual judge LLMs (e.g., Qwen-7B, Aya-23).
- **Static Corpus:** Medical guidelines are updated periodically. The current corpus does not include automatic refresh mechanisms. A production system would require periodic re-indexing.
- **Recommended V5 Configuration:** Based on ablation findings, a V5 combining recursive 500-char chunking with semantic retrieval and CrossEncoder re-ranking (without BM25) is predicted to achieve faithfulness >92% and relevancy >75%, and should be evaluated in future work.

12. References

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